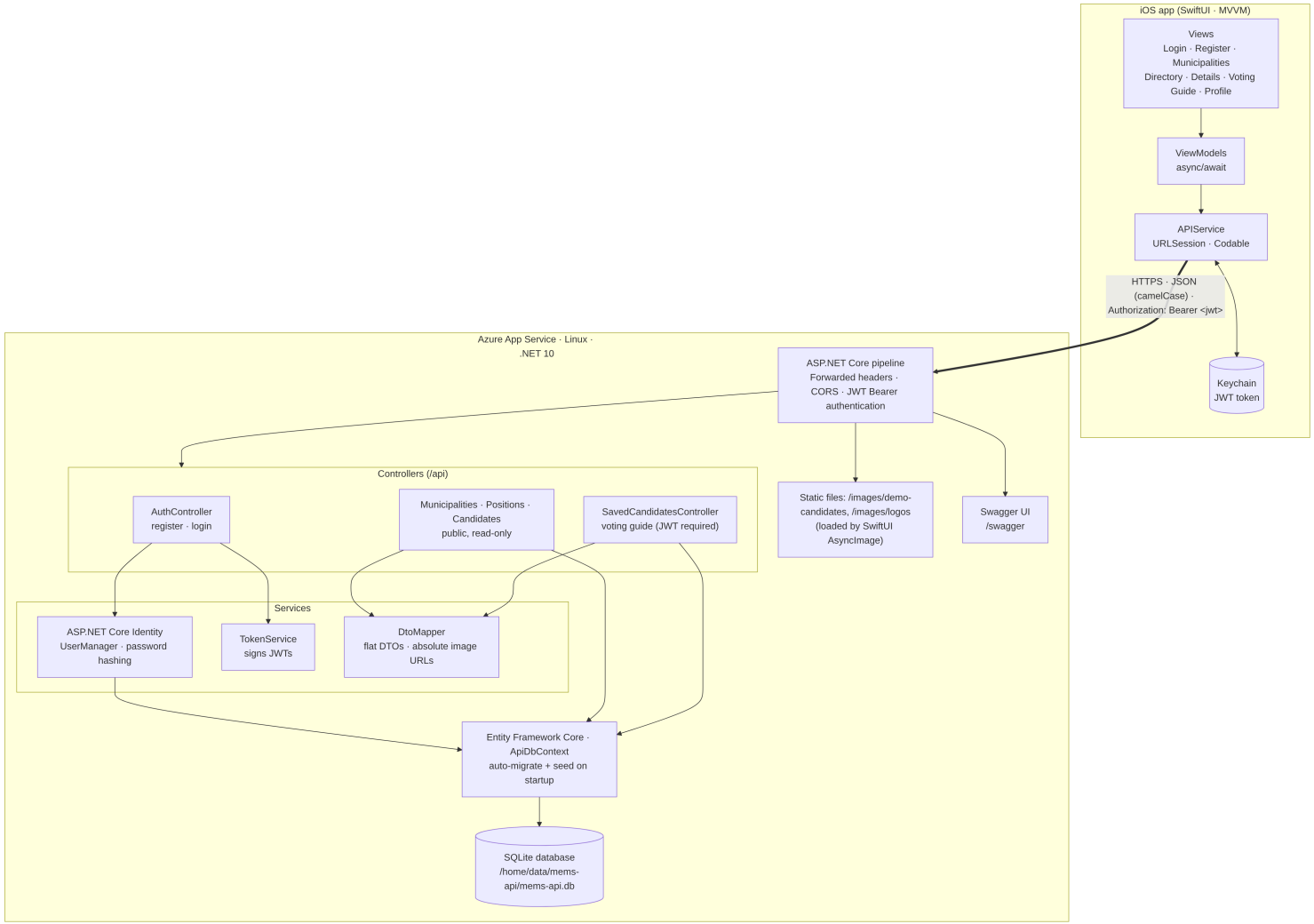
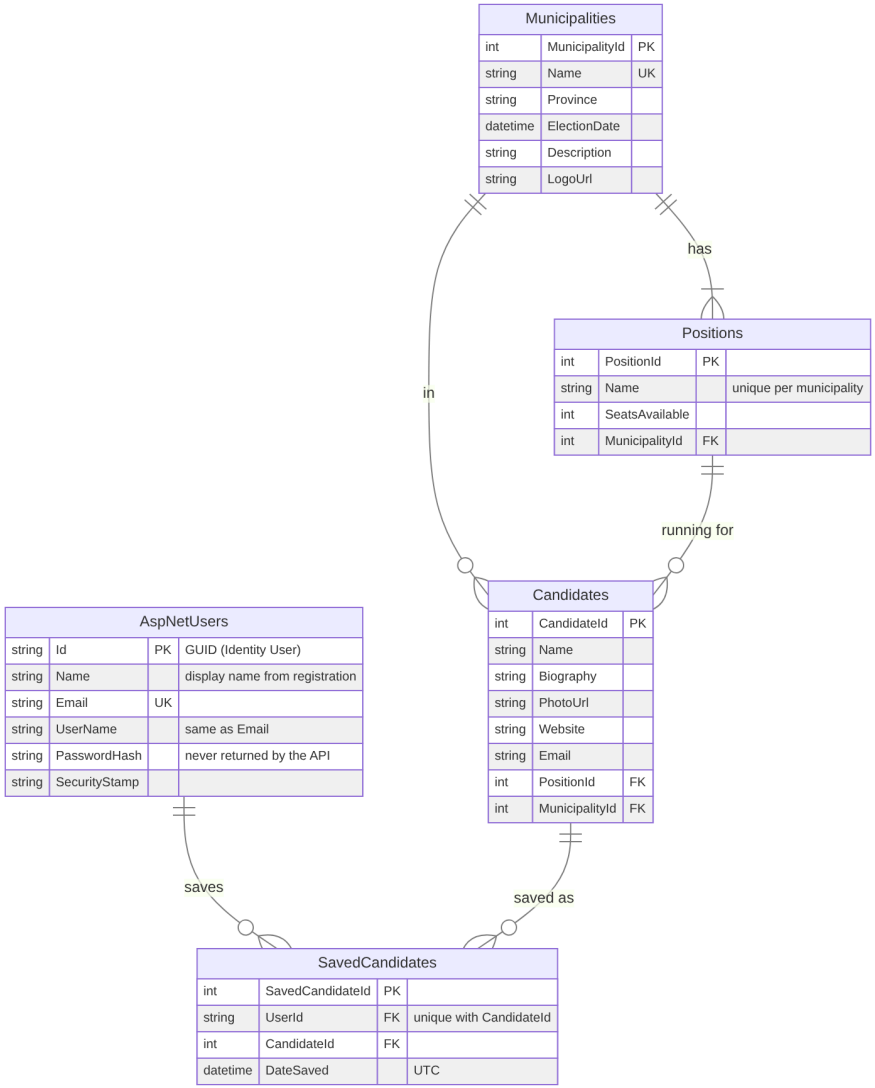
Municipal Elections Companion API

COMP4977 Assignment 1, Part 1 · ASP.NET Core Web API (.NET 10) · EF Core + SQLite · ASP.NET Core Identity · JWT
Azure API URL: https://mems-api-afisher-ane9a7cmfpb9b9at.canadacentral-01.azurewebsites.net
Swagger UI: https://mems-api-afisher-ane9a7cmfpb9b9at.canadacentral-01.azurewebsites.net/swagger
GitHub: https://github.com/AlexanderFisher-tech/MEMS-Assignment-1-main

Architecture Diagram



Database ERD



AspNetUsers is the ASP.NET Core Identity user table (extended with Name). Deleting a user or candidate cascades to their SavedCandidates; Municipality → Position → Candidate deletes are restricted. (UserId, CandidateId) is unique.

API Endpoints

All responses are camelCase JSON; errors are `{ "message": "..." }`. JWT endpoints require `Authorization: Bearer <token>` from `/api/auth/login`. Image URLs (photoUrl, logoUrl) are absolute HTTPS URLs.

Method	Endpoint	Auth	Request body	Success	Errors
POST	`/api/auth/register`	Public	`{ name, email, password }`	200 { message }	400
POST	`/api/auth/login`	Public	`{ email, password }`	200 { token, name, email }	400, 401
GET	`/api/municipalities`	Public		200 Municipality[]	
GET	`/api/municipalities/{id}`	Public		200 Municipality	404
GET	`/api/positions`	Public		200 Position[]	
GET	`/api/positions/{id}`	Public		200 Position	404
GET	`/api/candidates`	Public		200 Candidate[]	
GET	`/api/candidates/{id}`	Public		200 Candidate	404
GET	`/api/candidates/search?name={query}`	Public		200 Candidate[] (case-insensitive partial match)	
GET	`/api/savedcandidates`	JWT		200 SavedCandidate[] (current user)	401
POST	`/api/savedcandidates`	JWT	`{ candidateId }`	201 SavedCandidate	400, 401, 404, 409
DELETE	`/api/savedcandidates/{id}`	JWT		204 (by SavedCandidateId or CandidateId)	401, 404

Authentication Flow

